

Five Audits Under One Consumer Discipline: Renormalization, Quantum Darwinism, Lattice Hydrodynamics, Fluctuation Theorems, and Gauge Redundancy in Exact Finite Models

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This campaign models observation as consumption. A consumer reads a hidden substrate through a declared channel, spends a declared budget, and acts on a declared task, and every claim is settled by exact measurement in a declared finite model against bars fixed before the run. This paper applies that one discipline to five bodies of physics and reports twenty-seven experiment runs. In renormalization, whether a coarse model represents the substrate is set by the declared blocking rule, one rule keeps the coarse family exact while another mispredicts a declared correlator by 260 times its bar on the same substrate, the correlation length an observer reads is the one its channel's symmetry sector selects, and a declared perturbation's coarse image takes opposite signs under two blocking rules. In quantum Darwinism, redundant records exist exactly where the declared interaction wrote them, a globally scrambled state carries none, the order of interactions alone can hand one observer a record, equal-size fragments of one state differ in information by a factor of 36, and the rate at which two optimal observers agree runs from 0.510 to 1.000 as declared record strength grows, every disagreement computed from the declared channels. In lattice hydrodynamics, a reversible gas shows an entropy rise of 0.637 bits that retraces exactly under reversal, a shear mode freezes because of conservation laws the substrate actually has, the corrected substrate yields a viscosity of 0.842 that nine declared measurement prescriptions reproduce within 0.07 percent, and the decayed mode is recovered exactly by reversal while a single flipped bit reduces the recovered amplitude to 0.018 of its value. In fluctuation theorems, the Jarzynski and Crooks identities hold to machine precision path by path, restricted observers break them by exactly the amounts their declared detector models predict, correct inference through the true detector model restores them to machine precision while a misdeclared model leaves a larger defect than no correction at all, and the feedback identity holds with the acting controller's information and fails with a bystander's. In gauge theory, three routes to a Wilson loop agree to every printed digit, every gauge-fixed observer's link reading equals a gauge-invariant loop selected by its declared fixing, breaking the constraint is first-order visible in the fiber sector while the invariant sector responds at fourth order with measured slope 4.000003, and a coupled matter field has no bare identity apart from the observer's fiber coordinates. Six of the twenty-seven runs failed their declared bars. Each failure is recorded with its design error named, and each correction was run as a separate registration. Nothing in this paper is a claim about any laboratory system.

I. INTRODUCTION

This campaign models observation as consumption. A consumer reads a hidden substrate through a declared channel, spends a declared budget on the read, and acts on a declared task. Earlier stages measured what such consumers can distinguish in quantum models [1], what entropy they observe in reversible classical models [18], how declared prescriptions shape readings of rewriting systems [16], how declared constraints source geometry [17], and how the same structure plays out in games and decisions [15]. The present paper is the arc that carries the discipline across five established bodies of physics. The renormalization group of Kadanoff and Wilson [2–4], the quantum Darwinism program of Zurek [5, 6], the lattice gases of Hardy, de Pazzis, and Pomeau and of Frisch, Hasslacher, and Pomeau [8, 9], the fluctuation theorems of Jarzynski, Crooks, and Sagawa and Ueda [10–12], and the lattice gauge theory of Wegner and Elitzur

[13, 14] each already contain an observer, a blocking rule, a fragment, a coarse cell, a work meter, or a gauge fixing. The question asked here is uniform. Which parts of each standard story are properties of the substrate, and which are properties of the declared consumer.

The discipline is the same in all five tracks. Channels, tasks, environments, budgets, seeds, and falsification bars are declared in a track document before the runs they govern. Verdicts are computed from the declared items, never written first. Every result-bearing run writes an append-only evidence record, and every number in this paper is read from those committed records. When a run fails its declared bar, the failed verdict is retained unchanged, the design error is named, and the corrected declaration is run afterward as a separate registration. Six of the twenty-seven runs reported here failed, and Sec. VII presents all six with their named errors, because the sequence of verdicts is itself part of the evidence.

Three claim types run through the paper and are never conflated. Instrument claims state that the declared exact machinery passes closed-form controls, and each track opens with one. Structural claims state that a

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named piece of standard physics is, in the declared finite model, a property of the declared consumer rather than of the substrate, or measurably is not, and they carry the campaign's exploratory label. Replication claims state that a published exact result is reproduced in a declared finite instance. Nothing in this paper is a claim about any laboratory system. Every result is an exact measurement in a declared finite model, and no run is labeled as supporting more than that.

The five tracks return one coherent conclusion. What an observer reads is set by its declared structure. Representability, correlation length, and relevance follow the blocking rule. Redundancy and inter-observer agreement follow the declared interaction. Apparent dissipation follows the consumer's reversal reach. Fluctuation-identity defects follow the declared detector model. Gauge-variant readings follow the declared fixing. The apparent physics of coarse observers, in these models, is lawful consumer arithmetic.

II. COARSE REPRESENTABILITY, CORRELATION LENGTH, AND RELEVANCE FOLLOW THE DECLARED CHANNEL

The renormalization track asks which parts of the block-spin story belong to the substrate and which to the declared blocking channel. Its instrument layer, RG-0, works on the one-dimensional Ising ring, exact by transfer matrix and by full enumeration. Decimation of every other spin of a 12-ring at coupling $K = 0.5$ reproduces the 6-ring Ising distribution at the closed-form coupling

$$K' = \operatorname{artanh}(\tanh^2 K), \quad (1)$$

here $K' = 0.21689$, with maximum deviation 4.2×10^{-17} in every configuration probability, the correlation length halves exactly, and two decimations equal one decimation at the composed coupling to 2.1×10^{-17} . The channel-dependence control is the seed of the track. On a 12-ring at $K = 0.6$, majority-rule blocks of three produce a coarse distribution whose best nearest-neighbor Ising fit mispredicts the coarse next-nearest correlator by 0.0260, which is 260 times the declared 10^{-4} bar, while the decimation channel's full-distribution residual on the same substrate is 4.2×10^{-17} . Whether the microscopic model is representable in the coarse family is decided by the declared channel, not by the model.

RG-1 measures which long-distance quantity survives a declared family of channels. The substrate is the two-dimensional Ising model on a width-4 periodic cylinder at $K = 0.3$, exact through the 16×16 column transfer matrix, and the four declared column functionals are decimation to the first spin, majority vote, the adjacent pair product, and the column parity. The transfer matrix commutes with the global spin flip, so each channel's correlator is a sum over the eigenvectors its output couples to. The measured cross-sector amplitude leakage is exactly zero for all four channels, so each channel's decay rate is exactly its

sector's leading eigenvalue ratio. The odd-sector correlation length is 1.5026 and the even-sector length is 0.5194, a factor 2.893 apart, and the channel decides which one the observer reads. The recorded verdict of the first run is FAIL. Its route-agreement bar compared the eigenvector sum against direct transfer-matrix powers that subtracted order-one means from order-one moments, and at separation 8 the connected even-channel correlator is near 5×10^{-8} , so the declared comparison was numerically impossible as written and missed its bar at relative deviation 0.0386. The redeclared run, RG-1b, applies the mean subtraction before the matrix powers, a cancellation-free computation of the same quantity, and the two routes agree to 1.3×10^{-14} at every measured separation with every physics item unchanged. Correlation length in this model is consumer-selected, not consumer-invented, and the sectors themselves are the substrate's.

RG-2 transplants the campaign's budget flip, the measured accuracy difference between a coarse-graining chosen for the declared task and one chosen for fidelity at the same budget. The budget is one bit, a deterministic bipartition of the sixteen column states with all 32767 bipartitions enumerated, and the task is predicting the sign of a distant column's magnetization. The fidelity-optimal bipartition maximizes the compressed marginal's entropy. A declared theorem fixes the task optimum, and the exhaustive search and the theorem certify each other exactly in both runs. The first run's recorded verdict is FAIL because its declared setting, $K = 0.3$ at separation 4, has almost nothing to predict. The task-optimal accuracy measured 0.6000482 against a predict-the-majority baseline of nearly the same value, and the fidelity criterion's 840-member tie set contained a partition matching it exactly, so the flip was exactly zero. A flip cannot be measured where there is no signal. The redeclared run, RG-2b, moves to $K = 0.6$ at separation 2, where the distant sign is genuinely predictable, and the flip is positive, task-optimal accuracy 0.95766 against 0.95754 for the best member of its 840-way fidelity tie. The margin is 1.12×10^{-4} against the declared 10^{-4} bar, a pass with little room, so in this setting the flip is present but thin. The task-optimal partition assigns fifteen of the sixteen column states against the raw leading-eigenvector sign, the finite-distance correction the protocol asked about.

RG-3 is the universality gate. Three declared microscopic models are flowed under declared channels, and the family residual of a coarse distribution is its total-variation distance from the best nearest-neighbor Ising fit. The nearest-neighbor model at $K = 0.7$ flows exactly in-family under two decimations, with residuals below 7×10^{-16} . The model with an added next-nearest coupling flows toward the family, residual 0.157 to 0.032 to 0.004, so the extra coupling is measured irrelevant under decimation. Symmetry classes do not mix, the field model keeps a coarse magnetization of 0.777 while the symmetric models hold zero within 10^{-15} . The first run's recorded verdict is FAIL on its channel-relativity item, and the failure corrects the declaration's picture of the flow. The

TABLE I. RG-4 response multipliers, the induced coarse parameter over the microscopic perturbation, from the committed evidence record. The base model is the nearest-neighbor Ising 12-ring at $K = 0.6$ with perturbation strength 10^{-4} , and linearity holds with relative deviation 9.1×10^{-5} . The two declared channels disagree by 24 percent on the field's growth rate and disagree in sign on the next-nearest coupling.

Perturbation	decimation	majority
field	+1.834	+2.407
next-nearest coupling	+0.174	-0.071

declaration predicted that majority blocking would push the next-nearest model's residual up, and it fell, 0.161 to 0.0136, because from an out-of-family start the flow toward the trivial fixed point dominates any channel's family violation, and majority carries the substrate further down that flow than decimation does. The redeclared run, RG-3b, makes the corrected comparison from an exactly-in-family start. There one decimation leaves the family residual at 1.3×10^{-16} while one majority-of-three blocking raises it to 0.0117, so the channel decides whether the family survives at all, while for the out-of-family substrate both channels ride the dominant flow. Universality in this model is real but channel-shaped. What converges is convergence itself, and which family it converges through is the observer's choice.

RG-4 measures relevance itself as a channel property. Table I shows the four response multipliers. The field is relevant under both declared channels but with different eigenvalues, 1.834 under decimation against 2.407 under majority. The next-nearest coupling's multiplier is +0.174 under decimation and -0.071 under majority. The perturbation's coarse image differs in sign across the declared channels, so the direction a perturbation appears to flow is a property of the observer's blocking rule, not of the perturbation.

III. REDUNDANCY IS DECLARED OR ABSENT AND AGREEMENT BETWEEN OBSERVERS TRACKS RECORD STRENGTH

Quantum Darwinism explains agreement between observers by redundancy, many environment fragments carrying the same record of the system [5, 6]. That is intrinsically a statement about a family of consumers, each reading its own fragment. The track works on one system qubit and six environment qubits with exact density matrices, and its instrument layer, QD-0, passes four closed-form controls. A product state carries mutual information 3.2×10^{-16} . The GHZ branching state gives the classical plateau, one bit for every proper fragment and two bits for the whole environment, to 2.2×10^{-16} . A branching state with per-qubit record overlap $c = 1/2$ matches the closed form

$$I(S:F_f) = H\left(\frac{1+c^6}{2}\right) + H\left(\frac{1+c^f}{2}\right) - H\left(\frac{1+c^{6-f}}{2}\right) \quad (2)$$

bits at every fragment size f , with H the binary entropy, to 2.2×10^{-16} . The control that frames the track is a seeded Haar-random pure state of all seven qubits. Its mean single-qubit fragment information is 0.051 bits against a system entropy of 0.999 bits, while the full environment carries exactly twice the system entropy. Records are structure, not a generic feature of purity.

QD-1 is the emergence gate. Controlled rotations of declared angle θ from system to each environment qubit produce branching states whose fragment informations match Eq. (2) with overlap $c = \cos(\theta/2)$ and nothing fitted, measured to 2.7×10^{-15} at every angle on the ladder from $\pi/16$ to π . Redundancy, the number of disjoint single-qubit fragments reaching ninety percent of the system entropy, is 0 at every angle below π and 6 at π . The recorded verdict of the first run is FAIL on its basis-free control, and the failure is itself a measured finding. The control applied six sequential pairwise Haar-random unitaries, system with each environment qubit in turn, and ended with system entropy 0.665 bits and redundancy 1 rather than the declared 0. The last-touched environment qubit carries a redundancy-grade record, because the causal order of the interactions is itself declared structure. A sequence is structure, and an observer can be handed a record by the schedule of interactions alone. The redeclared run, QD-1b, replaces the control with a globally scrambled seeded Haar state with no interaction schedule at all, and it carries system entropy 0.9945 bits with redundancy 0. Redundancy in this model is declared or absent. The plateau appears exactly when and exactly as the declared interaction's basis dictates, and global scrambling without a declared basis carries no redundancy at all.

QD-2 measures the consumer's budget. For the symmetric consumer, equal angles on all six qubits, the smallest sufficient fragment size is 3 at every one of sixteen grid angles, and the closed form says why. A fragment holding half the environment has record overlap equal to its complement's, so its information equals the system entropy exactly at any nonzero angle. The portfolio consumer has declared unequal angles, so a fragment's overlap is the product over the qubits it holds and fragments of equal size are no longer equivalent. All 62 proper environment subsets match their closed forms to 1.1×10^{-15} , and the best and worst two-qubit fragments carry 1.612 and 0.045 bits, a factor of 36.0. Two consumers with identical fragment counts and identical access sizes read the same state with a thirty-six-fold information disparity. A budget is a portfolio, not a number.

QD-3 makes agreement itself the measured object. Two consumers hold disjoint two-qubit fragments of the QD-1 branching states and each performs its declared optimal two-outcome measurement, the Helstrom discrimination of its fragment's branch states. Each marginal success probability matches the Helstrom closed form [7] to 8.9×10^{-16} at every angle, and the joint outcome distribution computed on the full state and on the reduced four-qubit state agree to 6.7×10^{-16} . Figure 1 shows the

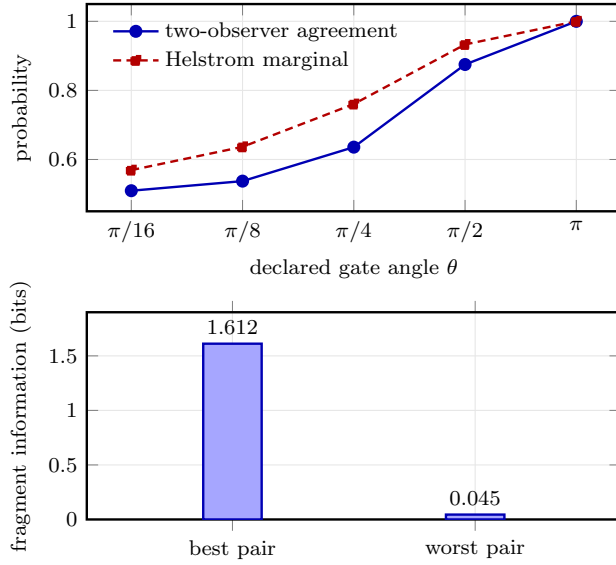


FIG. 1. Agreement and the portfolio in the Darwinism track. The upper panel shows the QD-3 probability that two optimal observers of disjoint two-qubit fragments report the same outcome, rising from 0.510 to 1.000 along the declared angle ladder, together with each observer's Helstrom success probability, which matches its closed form to 8.9×10^{-16} . The lower panel shows the QD-2 portfolio consumer, whose best and worst two-qubit fragments of the same state carry 1.612 and 0.045 bits, a factor of 36. Every point is read from the committed evidence records.

agreement curve, 0.510, 0.537, 0.636, 0.875, 1.000 along the angle ladder, barely above a coin flip at the weakest record and 1.000 at the perfect-record point. Two optimal observers agree exactly to the extent the declared interaction wrote records for them. Objectivity between observers, in this model, is record strength made mechanical, and every disagreement rate is the declared channels' Born arithmetic, never noise.

IV. SHEAR FREEZES ON HPP, VISCOSITY ON FHP IS PRESCRIPTION-INVARIANT, AND DISSIPATION IS THE CONSUMER'S REVERSAL REACH

The hydrodynamics track asks whether a transport coefficient belongs to the substrate or to the declared coarse-graining, and whether the second-law appearance at coarse scale is entirely the observer's. The instrument layer, HD-0, runs the HPP lattice gas [8] on a 16×16 torus. Mass 194 and both momentum components are integer-exact at every step, double velocity reversal reproduces the initial state bit for bit, and rotating the initial state commutes with evolution bit for bit. The declared coarse observer's entropy rises 0.637 bits from the ordered start, and on the reversed run the entropy curve retraces the forward curve with maximum deviation exactly zero. The rise is the observer's. The substrate forgets nothing.

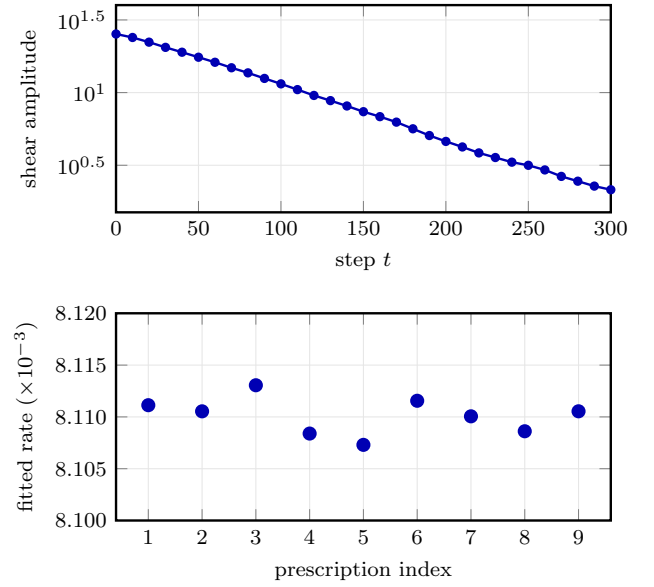


FIG. 2. The FHP shear mode and the prescription gate. The upper panel shows the HD-3 forward shear amplitude falling from 25.31 to 2.15 over 300 steps, the viscous decay of HD-2 reproduced on a new seed. The declared exact inverse returns every one of these numbers, the inverse-run sequence equals the forward sequence reversed with maximum deviation exactly zero. The lower panel shows the nine declared coarse-graining prescriptions of HD-2 applied to the same stored momentum series, cell sizes 1, 4, and 8 columns crossed with sampling cadences 1, 2, and 4 steps. The fitted rates span 0.008107 to 0.008113, a spread of 0.07 percent of the mean against the declared ten percent bar. Every point is read from the committed evidence records.

HD-1 aims a shear and a sound wave at a 64-torus ensemble, and its recorded verdict is FAIL on the shear items. The failure is the substrate's own physics, exactly measured. The shear amplitude did not decay at all, decay ratio exactly 1.0 at both wavenumbers with fitted rates at the 10^{-18} floating-point floor. The head-on collision exchanges a north-south pair for an east-west pair and streaming keeps each mover on its line, so the transverse momentum of every column and the longitudinal momentum of every row are separately conserved, and a shear mode can never relax. These are the spurious conservation laws of HPP, the pathology that motivated the FHP model [9], here measured by the declared instrument rather than assumed. The sound sector passed, with measured sound speed 1.008 and 1.012 of the lattice prediction and dispersion ratio 2.008. The redeclared run, HD-1b, converts the failure into exact bars. The per-column transverse momentum vector and the per-row longitudinal momentum vector are integer-exact constant at every one of 400 steps across the full ensemble, and the diffusive-scaling bar is withdrawn as unmeasurable on this substrate. The viscosity question moves to a substrate without the spurious invariants.

HD-2 declares the FHP-I hexagonal gas on a 64×64

triangular torus and validates it with exact geometry self-tests before any physics is read. The shear that HPP froze relaxes on FHP. The $k = 1$ amplitude falls to 0.170 of its window start and the $k = 2$ amplitude to 0.027, with rates 0.008114 and 0.036818 and a rate ratio of 4.538 inside the declared diffusive band. The measured viscosity is 0.842 at $k = 1$ and 0.955 at $k = 2$, beside the FHP-I Boltzmann estimate 0.742, a comparison and never a bar. Nine declared prescriptions, three cell sizes crossed with three sampling cadences, applied to the same stored momentum series, give fitted rates from 0.008107 to 0.008113, a spread of 0.07 percent of the mean against the declared ten percent bar (Fig. 2). The campaign's rewriting-system track found a growth rule whose measured dimension moved by 18 percent when the bookkeeping prescription changed [16]. The contrast is the finding. An exponential hydrodynamic mode's rate survives every declared linear cell average and every subsampling cadence, so which observables a prescription can touch is itself a measured distinction between substrates.

HD-3 is the dissipation gate. An FHP-I update is a bijection, so the exact inverse step is declared, and the forward shear ensemble is run 300 steps and then inverted. The initial state is recovered bit for bit in all 50 realizations. The projected shear amplitude fell from 25.31 to 2.15 on the way out, ratio 0.085, and the inverse-run amplitude sequence equals the forward sequence reversed with maximum deviation exactly zero, float for float (Fig. 2). Every dissipated number came back, so no strict dissipative law governs the coarse observable, and the decay is the observer's accounting. One declared bit flipped at the turnaround destroys it. The perturbed inverse returns its realization at amplitude 0.437 against a true initial 24.04, a return ratio of 0.018 against the declared bar of one half, while the 49 untouched realizations return bit for bit, a control inside the item. The Hamming distance between the perturbed and true inverse trajectories floods from 1 bit to 3047 by 30 steps and 11007 by 60, then sits near 11200, close to 11182, the expected Hamming distance between two independent configurations at the declared occupancy on the realization's 24576 bits. Dissipation on this substrate is declared or absent in the consumer. The coarse mode's decay is recoverable by any observer with exact reversal reach and irrecoverable after the loss of one bit.

V. RESTRICTED OBSERVERS BREAK THE FLUCTUATION IDENTITIES BY EXACTLY THEIR DECLARED CHANNELS

The Jarzynski equality [10] and the Crooks relation [11] are exact identities for an observer who sees the full trajectory. The track asks what happens to them for a consumer with a restricted channel. The instrument layer, FT-0, declares a three-state system with a linear energy protocol over eight steps and Metropolis kernels in detailed balance, and enumerates the full path space

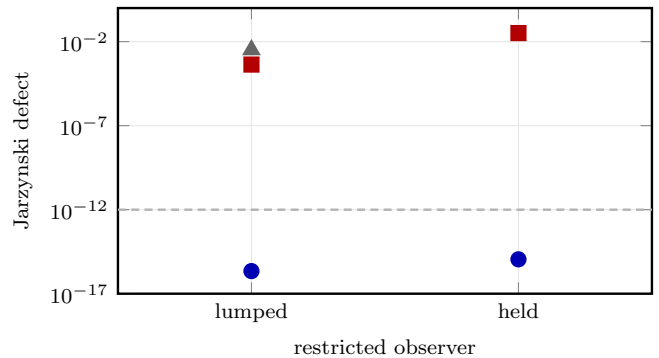


FIG. 3. The defect ladder and its restoration. Squares mark the naive apparent-work defects, 4.33×10^{-4} for the lumped observer and 0.0324 for the held observer, while the complete and blind observers' defects are exactly zero and sit off this logarithmic scale. Circles mark the Bayes work estimates filtered through the true declared detector model, which restore both identities below the dashed declared bar of 10^{-12} , to 2.2×10^{-16} and 1.1×10^{-15} . The triangle marks the same filter built from a misdeclared model, kernels at inverse temperature 0.9, which leaves the lumped observer at 3.27×10^{-3} , a larger defect than no correction at all. Every point is read from the committed evidence records.

of 19683 paths exactly. The Jarzynski average

$$\langle e^{-W} \rangle = e^{-\Delta F} \quad (3)$$

holds with deviation exactly zero, and the Crooks relation holds path by path, not only on average, with maximum deviation 2.2×10^{-19} over all 19683 paths. Mean work is -0.0788 against $\Delta F = -0.1021$, a dissipation of 0.0233, and the frozen protocol yields exactly zero work on every path. This is the replication half of the track, the published identities reproduced at the finest grain the model has.

FT-1 is the lawful-demon audit. Four declared observers read the same path space. The complete observer sees the full path and its defect is exactly zero. The blind observer sees nothing, books the equilibrium free-energy change as its work, and its defect is also exactly zero, because total blindness satisfies the identity trivially. The lumped observer, which merges the two lowest states, has defect 4.33×10^{-4} , and the held observer, which reads the state only at even stages, has defect 0.0324. The defect lives strictly between full sight and total blindness. The lawfulness clause is the route agreement. For both restricted observers, the defect computed by full path enumeration equals the defect computed independently from the exact hidden-Markov marginal over their observed records, to 6.7×10^{-16} . No apparent violation is an anomaly. Each is the declared detector model's exact arithmetic.

FT-2 asks whether the identity can be restored. The declared restoration replaces the naive apparent work with the Bayes work estimate, minus the log of the conditional expectation of e^{-W} given the observer's record, computed by exact filtering through the declared detector model.

With the true model the restored identity holds at 2.2×10^{-16} for the lumped observer and 1.1×10^{-15} for the held observer. With a misdeclared filter, the same machinery built from kernels at inverse temperature 0.9, the lumped observer's restored defect is 3.27×10^{-3} , larger than its naive defect of 4.33×10^{-4} (Fig. 3). Confident wrong inference is worse than no inference. The fluctuation identity for a budgeted consumer is an inference theorem, restored exactly by lawful inference through the true declared channel and by nothing less.

FT-3 replicates the Sagawa-Ueda feedback generalization [12] and then asks whose information the identity contains. A controller measures the state mid-protocol through a declared noisy channel with error rate 0.2 and switches the remaining protocol on one outcome. The identity

$$\langle e^{-(W-\Delta F)-I} \rangle = 1 \quad (4)$$

holds to 5.6×10^{-15} with the controller's pointwise information. The bare identity without the information term fails by 7.3×10^{-3} , efficacy 0.9927, the feedback demon's measured advantage. Refeeding Eq. (4) with the pointwise information of a declared bystander who sees the same measurement through additional noise, total error rate 0.4, leaves a defect of 5.3×10^{-3} , while the controller's own information restores exactness. The identity knows which observer acts. The information-corrected second law holds with margin 0.489 nats. The information that appears in fluctuation identities, in this model, is the acting consumer's, not anyone's.

VI. GAUGE-VARIANT READINGS ARE INVARIANT LOOPS IN THE OBSERVER'S COORDINATES

A gauge orbit is the fiber of a projection, many link configurations behind one physical state. The track asks whether gauge symmetry is anything but the fiber structure of the observer's projection. The instrument layer, GG-0, enumerates all 2^{18} configurations of $\mathbb{Z}_2$ gauge theory [13] on a 3×3 periodic lattice at $K = 0.4$. Every sampled orbit has size exactly 256 with the action exactly constant on it. The single-plaquette Wilson expectation agrees across the full enumeration, a tree-gauge-fixed enumeration, and the exact torus closed form

$$\langle W_P \rangle = \frac{t + t^8}{1 + t^9}, \quad t = \tanh K, \quad (5)$$

to every printed digit, 0.38032051536830963 on all three routes. Every single-link expectation in the full ensemble vanishes exactly, Elitzur's theorem [14] at machine precision, while the tree-gauge-fixed observer's largest non-tree link reading is 0.3803, the single-plaquette expectation itself. In tree gauge a non-tree link equals a Wilson loop through the tree, so the observer's gauge-variant bookkeeping is a gauge-invariant quantity wearing a gauge-variant label.

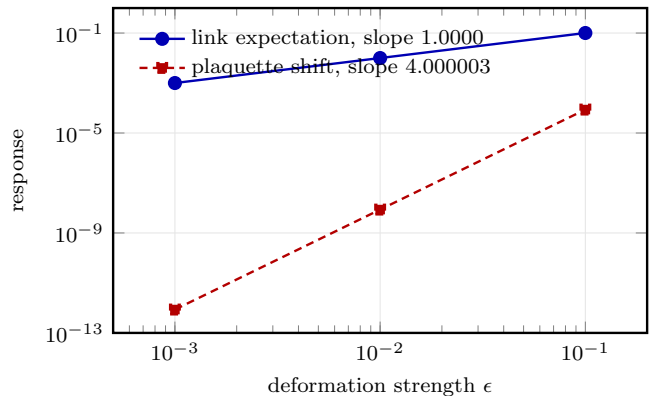


FIG. 4. The GG-2b sector split under the declared link deformation. The maximum link expectation responds at first order, measured log-log slope 1.0000, visible at machine precision at every positive ϵ . The plaquette Wilson expectation's shift from its undeformed value responds at fourth order, measured slope 4.000003, because no two links on this lattice close a loop with a plaquette, so the second-order coefficient vanishes identically by loop parity. Both series are read from the committed evidence record, the plaquette shift as the difference of recorded expectations.

GG-1 makes that mechanism the measured object across a declared family of three spanning-tree observers. Every gauge-invariant observable agrees between the full ensemble and every observer's slice with deviation exactly zero. The detector model is exact. Every non-tree link reading of every observer equals the full-ensemble expectation of the closed loop formed by that link and its tree path, deviation exactly zero, contractible loops match the area closed form, and all 18 winding loops found by the GF(2) certificate read exactly zero. The observers measurably disagree, maximum pairwise link spread 0.855, while the full-ensemble Elitzur zero stands. Disagreement between observers is exactly the difference in which invariant loops their trees select, and nothing any observer reads is outside the invariant algebra.

GG-2 replays the campaign's declared-or-absent constraint gate in gauge language. The link ensemble is deformed by a gauge-breaking link field of strength ϵ on a declared ladder, all 2^{18} configurations enumerated at each point. At $\epsilon = 0$ the zeros are exact. At every positive ϵ the maximum link expectation responds at first order, measured slope 1.0000, and the deformed action is strictly non-constant on every sampled orbit. The broken constraint is visible at machine precision immediately, not asymptotically. The recorded verdict of the first run is FAIL on the invariant sector's window. The declaration assumed a generic quadratic response and declared the window [1.5, 2.5] for the plaquette shift's log-log slope, and the measured slope is 4.000003. The declaration underestimated the substrate's silence. At second order the deformation enters through pairs of links, no two links on this lattice close a loop with a plaquette, so the second-order coefficient vanishes identically and the

first surviving response is quartic, four deformation links closing a loop. The redeclared run, GG-2b, carries that loop-parity argument as its declared prediction with window [3.5, 4.5], and passes (Fig. 4). Constraint structure in this model is an isolated algebraic point, declared or absent, with no approximately-gauge ensemble in between, and the observer that watches only invariants is fourth-order blind to a breaking the fiber sector announces at first order.

GG-3 couples $\mathbb{Z}_2$ matter to the links, $K = 0.4$ and $J = 0.1$. The unitary-gauge identity, absorbing the matter into the links, maps the coupled model onto the GG-2 deformed ensemble at $\epsilon = J$, verified on a fully enumerated 2×2 torus where the partition functions and every dressed correlator agree between the two routes with deviation exactly zero, so the 3×3 coupled theory is computed exactly through the link enumeration. Elitzur for matter holds. The bare two-point matter correlator without its connecting line is exactly zero, while the dressed correlator, endpoints joined by the link line, reads 0.1008 on the small torus and 0.1004 on the large one. Charge exists only as the endpoint of a line. The observer family splits exactly as declared. The unitary-gauge observer reads bare matter as exactly one, the tree-gauge observer reads 0.101, an observer spread of 0.899, while the dressed observables agree across both with deviation exactly zero. What an observer calls the bare matter field is its fiber coordinate, nothing more.

VII. SIX RUNS FAILED THEIR DECLARED BARS AND EACH FAILURE HAS A NAME

Six of the twenty-seven runs recorded verdict FAIL, every failure was located and named, and every correction was run as a separate registration with the original record retained unchanged. The campaign's previous paper followed the same pattern for its two failures, a residual threshold where a proof was needed and a double-counted tie set [15]. The six failures of this arc each taught something specific.

RG-1 declared a route-agreement bar that subtracted order-one means from order-one moments after powering the transfer matrix, and at the declared separation the connected value is near 5×10^{-8} , so the comparison was numerically impossible as written. The correction subtracts the mean before powering. A declared bar must be computable at the scale it will be compared against.

RG-2 declared a prediction task in a setting with almost no signal, task accuracy 0.6000482 against a predict-the-majority baseline of nearly the same value, so the flip it was built to measure was exactly zero. The correction moves to a setting where the target is genuinely predictable. A flip cannot be measured where there is nothing to predict.

RG-3 predicted the direction of a channel comparison from an out-of-family start, and the substrate's own flow toward triviality dominated the channel difference. The

correction states the channel-relativity claim from an exactly-in-family start, where it is measured and stands. A channel effect must be isolated from the flow it rides on.

QD-1 declared a basis-free control that applied its scrambling unitaries in sequence, and the causal order imprinted a redundancy-grade record on the last-touched qubit. The correction uses a globally scrambled state with no schedule. A sequence is structure, and this failure is itself a finding about how observers come to hold records.

HD-1 declared a shear-decay bar on a substrate whose spurious conservation laws freeze shear exactly, decay ratio exactly one. The correction measures the spurious invariants as exact conservation laws and moves viscosity to a substrate without them. This failure is a replication of the pathology that motivated FHP [9], found by the declared instrument.

GG-2 declared a quadratic window for the invariant sector's response, and the measured slope is 4.000003 because loop parity closes the quadratic channel identically. The correction carries the loop-parity argument as its prediction. The declaration underestimated how silent the substrate's invariant sector is.

Three of the six failures, QD-1, HD-1, and GG-2, are measurements of real structure the declaration did not anticipate, and the other three are named bookkeeping errors in the declarations themselves. In all six the corrected redeclaration passed with the measured physics unchanged, and in all six the failed record is part of the evidence.

VIII. WHAT THIS PAPER ESTABLISHES

Tables II and III register the twenty-seven runs with their recorded verdicts. Read together they answer the arc's question inside its declared scope. In the renormalization models, whether a substrate is representable in a coarse family, which correlation length an observer reads, whether a coarse theory converges to a family, and how strongly a perturbation grows are all set by the declared channel, down to the sign of a response multiplier. In the Darwinism models, redundancy exists exactly where a declared interaction wrote it, budgets are portfolios rather than numbers, and agreement between observers is the declared record strength, never noise. In the lattice-gas models, the entropy rise, the frozen shear, the viscous decay, and the loss of recoverability are each exactly attributable, the rise to the coarse observer, the freeze to the substrate's own invariants, the coefficient to the substrate with every declared prescription agreeing, and the irreversibility to the consumer's reversal reach. In the fluctuation models, every identity defect of a restricted observer is its declared channel's arithmetic, restoration is an inference theorem, and the information term belongs to the acting consumer. In the gauge models, everything any gauge-fixed observer reads is an invariant loop its declared fixing selects, constraint structure is declared or

TABLE II. The ledger of the renormalization and Darwinism tracks. Verdicts are recorded exactly as computed from the declared items of each run, including the failures. Measured in model, exploratory label, means an exact measurement in a declared finite construction with append-only evidence records and pre-declared bars, seeds, and tolerances. Key numbers are quoted from the committed records.

Run	Verdict	Key measurements	Label
RG-0 instrument	PASS	decimation exact to 4.2×10^{-17} with the closed-form coupling flow, correlation length halves exactly, flow composes, majority-rule nearest-neighbor fit mispredicts the next-nearest correlator by 0.0260, 260 times its bar	instrument
RG-1 channel family	FAIL	cross-sector leakage exactly zero for all four channels, sector correlation lengths 1.5026 and 0.5194, route bar failed at 0.0386 by catastrophic cancellation in the declared comparison	measured in model, exploratory
RG-1b corrected route	PASS	cancellation-free route agrees to 1.3×10^{-14} , decay rate is exactly the declared channel's sector eigenvalue ratio, length ratio 2.893	measured in model, exploratory
RG-2 budget flip	FAIL	theorem and exhaustive search certify each other exactly, declared setting had no signal, accuracy 0.6000482, flip exactly zero	measured in model, exploratory
RG-2b corrected setting	PASS	flip positive at 1.12×10^{-4} against the 10^{-4} bar, a pass with little room, accuracies 0.95766 against 0.95754 over an 840-way fidelity tie	measured in model, exploratory
RG-3 universality gate	FAIL	in-family flow exact, out-of-family residual falls 0.157 to 0.032 to 0.004, majority reduced the residual 0.161 to 0.0136 where the declaration predicted an increase	measured in model, exploratory
RG-3b corrected start	PASS	from the in-family start decimation leaves residual 1.3×10^{-16} while majority raises it to 0.0117, the channel decides whether the family survives	measured in model, exploratory
RG-4 relevance	PASS	field multiplier 1.834 against 2.407 across channels, next-nearest multiplier $+0.174$ against -0.071 , channel-signed, linearity 9.1×10^{-5}	measured in model, exploratory
QD-0 instrument	PASS	GHZ plateau and partial-record closed form to 2.2×10^{-16} , Haar control 0.051 bits per fragment against system entropy 0.999	instrument
QD-1 emergence gate	FAIL	branching closed form to 2.7×10^{-15} with nothing fitted, sequential scrambling control carried redundancy 1, a recency record, itself a finding	measured in model, exploratory
QD-1b corrected control	PASS	global Haar state carries entropy 0.9945 bits with redundancy 0, plateau appears exactly as the declared basis dictates	measured in model, exploratory
QD-2 budget knob	PASS	smallest sufficient fragment 3 at every angle, half-environment information equals system entropy exactly, best against worst equal-size fragments differ by the factor 36.0	measured in model, exploratory
QD-3 agreement audit	PASS	Helstrom marginals to 8.9×10^{-16} , Born routes to 6.7×10^{-16} , agreement 0.510 to 1.000 along the record-strength ladder	measured in model, exploratory

absent with the invariant sector fourth-order blind to its breaking, and bare matter is a fiber coordinate.

These are structural results about declared finite models carrying the exploratory label of the campaign's evidence-grading scheme. They show that in these models the named pieces of standard physics are consumer arithmetic. They do not show that laboratory physics works this way, and no run in this paper can bear on that question.

IX. METHODS AND REPRODUCIBILITY

All runs follow the campaign's standing discipline. Substrates, channels, observables, seeds, tolerances, and bars were declared in the five track documents before the runs they govern, verdicts are computed

from the declared items, and every result-bearing run writes an append-only JSON evidence record containing the declared parameters, the measured outcomes, and a canonical SHA-256 hash of the record body. The records for this paper are `rg0-instrument.json` through `rg4-relevance.json`, `qd0-instrument.json` through `qd3-agreement.json`, `hd0-instrument.json` through `hd3-dissipation-gate.json`, `ft0-instrument.json` through `ft3-feedback.json`, and `gg0-instrument.json` through `gg3-minimal-coupling.json` in the campaign repository, twenty-seven records in all, with the committed runners alongside them. The failed RG-1, RG-2, RG-3, QD-1, HD-1, and GG-2 records are retained unchanged, and each corrected declaration is a separate record with its own registration, so the sequence of verdicts is itself

TABLE III. The ledger of the hydrodynamics, fluctuation, and gauge tracks, continuing Table II, with the scope row of the whole arc.

Run	Verdict	Key measurements	Label
HD-0 instrument	PASS	conservation integer-exact, double reversal bit-exact, entropy rise 0.637 bits retraced with deviation exactly zero	instrument
HD-1 transport	FAIL	shear decay ratio exactly 1.0, the spurious HPP invariants measured, sound speed 1.008 and 1.012 of prediction, dispersion ratio 2.008	measured in model, exploratory
HD-1b invariants exact	PASS	per-column and per-row momenta integer-exact over 400 steps and the full ensemble, diffusive bar withdrawn as unmeasurable	measured in model, exploratory
HD-2 FHP viscosity	PASS	viscosity 0.842 beside the Boltzmann comparison 0.742, rate ratio 4.538 in the diffusive band, nine prescriptions spread 0.07 percent against the ten percent bar	measured in model, exploratory
HD-3 dissipation gate	PASS	inverse bit-exact in 50 realizations, amplitude retrace deviation exactly zero, one flipped bit collapses the return to 0.018 with the Hamming lightcone flooding to full decorrelation	measured in model, exploratory
FT-0 instrument	PASS	Jarzynski deviation exactly zero, Crooks path by path to 2.2×10^{-19} over 19683 paths, dissipation 0.0233, frozen protocol exact	instrument, and replication
FT-1 lawful demon	PASS	defects, complete 0, lumped 4.33×10^{-4} , held 0.0324, blind 0, each restricted defect predicted by its detector model to 6.7×10^{-16}	measured in model, exploratory
FT-2 restoration	PASS	true-model Bayes estimate restores the identity to 2.2×10^{-16} and 1.1×10^{-15} , a misdeclared filter leaves 3.27×10^{-3} , worse than naive	measured in model, exploratory
FT-3 feedback	PASS	Sagawa-Ueda to 5.6×10^{-15} with the controller's information, bare identity off by 7.3×10^{-3} , bystander's information leaves 5.3×10^{-3}	replication, and measured in model, exploratory
GG-0 instrument	PASS	Wilson plaquette 0.38032051536830963 on three routes to every digit, Elitzur exactly zero, orbits size 256 with constant action	instrument, and replication
GG-1 observer family	PASS	every non-tree reading equals its tree-selected invariant loop with deviation exactly zero, 18 winding loops read exactly zero, observer spread 0.855	measured in model, exploratory
GG-2 Gauss replay	FAIL	link sector first-order loud, slope 1.0000, plaquette slope 4.000003 outside the declared quadratic window, loop parity closes the second order exactly	measured in model, exploratory
GG-2b corrected window	PASS	quartic window declared from the loop-parity argument, slope 4.000003 inside it, the invariant sector is fourth-order blind	measured in model, exploratory
GG-3 minimal coupling	PASS	unitary-gauge identity exact, bare matter correlator exactly zero against dressed 0.1008 and 0.1004, observer bare spread 0.899 with dressed agreement exact	measured in model, exploratory
Scope	non-claim	nothing in this paper is evidence about laboratory renormalization, decoherence experiments, physical fluids, thermodynamic experiments, or continuum gauge fields	non-claim

part of the evidence. Every number and every figure coordinate in this paper is read from those committed records. The figure tables are generated deterministically by a committed script, `make_figdata_gen2.py`, with no

free parameters, whose in-script assertions bind every emitted coordinate and every headline number quoted in the text to the record values it reads. A reducer may summarize records but may not rewrite them.

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